Expect the OA to be primarily coding + CS fundamentals, with some AI/GenAI awareness

Previous year programming & DSA

Question 1: Guess the Word

Problem:
Given N strings, find the first word having the maximum odd length. If no word has an odd length, print "Better luck next time".

Sample Test Case 1

Input:

5

Hello Good morning Welcome you

Output:

morning

Sample Test Case 2

Input:

3

Go to hell

Output:

Better luck next time

Question 2: Array Subarray

Problem:
Given an array and an integer k, consider every contiguous subarray of length k. Find the minimum element in each subarray and return the maximum among these minimums.

Sample Test Case 1

Input:

1

5

1

2

3

1

2

Output:

3

Sample Test Case 2

Input:

3

5

1

3

2

5

4

Output:

3

Question 3: Good Prime Number

Problem:
Given integers N and K, find the Kth number greater than N whose sum of digits is a prime number.

Sample Test Case 1

Input:

4

4

Output:

12

Sample Test Case 2

Input:

17

5

Output:

29

Question 4: Minimum Number of Swaps

Problem:
Given an array, arrange it in ascending order using the minimum number of swaps. Return the minimum number of swaps required.

Sample Test Case 1

Input:

4

4 3 1 2

Output:

2

Sample Test Case 2

Input:

5

1 5 4 3 2

Output:

2

Question 5: Seating Arrangement in Exam Hall

Problem:
Find the Nth binary number whose representation does not contain consecutive 1s.

The sequence starts as:

1 10 100 101 1000 1001 1010 10000 10001 10010 ...

Sample Test Case 1

Input:

3

4

6

9

Output:

101

1001

10001

Sample Test Case 2

Input:

3

1

2

5

Output:

1

10

1000

Question 6: Game of Clicks

Problem:
Given the lowest and highest TV channels, a list of blocked channels, and a sequence of channels to watch, find the minimum number of remote-control clicks required to visit all requested channels.

Sample Test Case 1

Input:

1

20

2

18

19

5

15

14

17

11

17

Output:

8

Question 7: Consecutive Prime Sum

Problem:
Given N, count the prime numbers less than or equal to N that can be represented as the sum of consecutive prime numbers starting with 2.

Sample Test Case 1

Input:

5

Output:

1

Sample Test Case 2

Input:

20

Output:

2

Question 8: Majority Element

Problem:
Given an array, find the element that occurs more than n/2 times. If no such element exists, return -1.

Sample Test Case 1

Input:

9

3 3 4 2 4 4 2 4 4

Output:

4

Sample Test Case 2

Input:

5

1 2 3 4 5

Output:

-1

Question 9: Smallest Window in a String

Problem:
Given two strings S and P, find the smallest substring of S that contains all characters of P, including duplicates. If no such substring exists, return -1.

Sample Test Case 1

Input:

timetopractice

toc

Output:

toprac

Sample Test Case 2

Input:

zoomlazapzo

oza

Output:

apzo

Sample Test Case 3

Input:

abc

xyz

Output:

-1

Question 10: Spiral Matrix

Problem:
Given a 2D matrix, traverse the matrix in spiral order.

Sample Test Case 1

Input:

5 4

1 2 3 4

5 6 7 8

9 10 11 12

13 14 15 16

17 18 19 20

Output:

1 2 3 4 8 12 16 20 19 18 17 13 9 5 6 7 11 15 14 10

Sample Test Case 2

Input:

3 3

1 2 3

4 5 6

7 8 9

Output:

1 2 3 6 9 8 7 4 5

AI / GenAI OA Practice Questions

Section A — LLM Fundamentals

Q1. What is a Large Language Model (LLM)?

A. A database designed to store documents
B. A machine-learning model trained to understand and generate language
C. A compiler for natural-language programs
D. A vector database

Q2. What is a token in an LLM?

A. A database record
B. A unit of text processed by the model
C. A type of neural network
D. A vector database entry

Q3. Why are tokens important for LLM applications?

A. They determine database indexing
B. They affect context limits and usage/cost
C. They replace embeddings
D. They eliminate hallucinations

Q4. What is a context window?

A. The number of users an LLM can support
B. The amount of input/output context the model can process within a request
C. The number of layers in the model
D. The size of the vector database

Q5. What does temperature generally control in an LLM?

A. Model training speed
B. Randomness/variability of generated responses
C. Context-window size
D. Number of tokens in the input

Q6. For a task requiring highly deterministic output, which temperature would generally be preferred?

A. Higher temperature
B. Lower temperature
C. Temperature has no effect
D. Random temperature

Section B — Prompt Engineering

Q7. What is prompt engineering?

A. Training a new LLM from scratch
B. Designing effective instructions/input to obtain desired model behavior
C. Creating a vector database
D. Compressing an LLM

Q8. What is the purpose of a system prompt?

A. Store embeddings
B. Provide high-level instructions/context governing the model's behavior
C. Train the model
D. Perform database queries

Q9. What is few-shot prompting?

A. Giving the model no examples
B. Providing a few examples in the prompt to guide the desired output
C. Fine-tuning the model
D. Increasing temperature

Q10. Which approach is generally best when you need the LLM to return predictable structured data?

A. Give vague instructions
B. Request a structured format/schema
C. Increase temperature
D. Remove context

Q11. A developer asks an LLM:

"Tell me about this."

The model gives inconsistent answers.

What is the best first improvement?

A. Increase temperature
B. Make the prompt more specific and provide relevant context
C. Delete the model
D. Increase database size

Section C — Hallucination

Q12. What is an LLM hallucination?

A. Model crashes
B. Model generates information that is incorrect, fabricated or unsupported
C. Model runs slowly
D. Model cannot tokenize text

Q13. Which approach can help reduce hallucinations in a knowledge-based application?

A. RAG
B. Increasing randomness
C. Removing context
D. Ignoring retrieved information

Q14. Why can an LLM hallucinate even when it produces a confident response?

A. Confidence in wording does not guarantee factual correctness
B. LLMs always access the internet
C. LLMs always query databases
D. Hallucinations occur only because of SQL

Section D — Embeddings

Q15. What is an embedding?

A. A text-generation algorithm
B. A numerical vector representation of data

C. A SQL query
D. An API endpoint

Q16. What is the primary purpose of text embeddings?

A. Convert text into numerical representations useful for similarity/search tasks
B. Generate passwords
C. Compress databases
D. Replace an LLM

Q17. Two documents have similar meanings but use different words.

Which technique can help identify their semantic similarity?**

A. Exact string matching
B. Embeddings
C. Sorting
D. Hashing only

Q18. Which metric is commonly used to compare embedding vectors for semantic similarity?

A. Cosine similarity
B. HTTP status
C. SQL JOIN
D. CPU utilization

Section E — Vector Databases

Q19. Why are vector databases used in GenAI applications?

A. To store source code only
B. To efficiently store and search vector embeddings
C. To train CPUs
D. To replace APIs

Q20. Which of the following is a vector database/vector search technology?

A. Pinecone
B. Microsoft Word
C. Git
D. Jenkins

Q21. What is similarity search?

A. Finding vectors that are mathematically/semantically close to a query vector
B. Finding identical usernames
C. Sorting database rows alphabetically
D. Searching only exact keywords

Q22. What is the relationship between embeddings and vector databases?

A. Embeddings provide vector representations; vector databases store/search those representations
B. They are exactly the same thing
C. Vector databases generate all LLM responses
D. Embeddings replace APIs

Section F — RAG

This is one of the **most important sections** because RAG, embeddings, vector databases and semantic search are explicitly mentioned in the JD.

Q23. What does RAG stand for?

A. Retrieval-Augmented Generation
B. Random AI Generation
C. Retrieval Analysis Gateway
D. Reinforcement-Augmented Graph

Q24. What is the primary purpose of RAG?

A. Give an LLM access to relevant external/contextual information during generation
B. Replace all databases

C. Train an LLM from scratch
D. Increase CPU speed

Q25. What is the correct basic RAG pipeline?

A.

LLM → Documents → Embeddings → User

B.

Documents → Chunks → Embeddings → Vector DB

↓

User Query → Retrieval → Relevant Context → LLM → Answer

C.

User → Database → Compiler

D.

Documents → LLM training → Delete documents

Q26. Why are documents divided into chunks in a RAG system?

A. To make retrieval more focused and manageable
B. To increase database duplication
C. To eliminate embeddings
D. To train the operating system

Q27. What happens during the retrieval stage of RAG?

A. The LLM is retrained
B. Relevant document chunks are retrieved based on the user's query
C. The database is deleted
D. The prompt is converted into machine code

Q28. What happens after relevant documents are retrieved?

A. They can be supplied as context to the LLM to generate the answer
B. They are automatically deleted
C. They replace the LLM
D. They become source code

Q29. RAG is particularly useful when:

A. Information is dynamic or domain-specific
B. No external knowledge is required
C. You want to remove all context
D. You don't need retrieval

Section G — RAG Troubleshooting

These are **very good interview/OA-style conceptual questions**.

Q30. A RAG system frequently retrieves irrelevant documents. What could be the reason?

A. Poor chunking
B. Poor embeddings
C. Incorrect retrieval configuration
D. All of the above

Q31. A RAG chatbot gives an incorrect answer even though the correct document exists in the vector database. What should you investigate first?

A. Whether the correct document was retrieved
B. Keyboard layout
C. CPU brand
D. Git branch name

Q32. A document is extremely large and retrieval produces poor results. What could improve the system?

A. Appropriate chunking
B. Remove embeddings

C. Increase random output
D. Delete the vector database

Section H — RAG vs Fine-Tuning

Q33. Which statement is correct?

A. RAG and fine-tuning are identical
B. RAG provides external context at inference time, while fine-tuning updates model behavior through additional training
C. Fine-tuning is simply SQL querying
D. RAG always modifies model weights

Q34. A company frequently updates its internal documentation. Which approach is generally attractive for answering questions over the latest documents?

A. RAG
B. Retraining the entire LLM after every document change
C. Removing the documents
D. Increasing temperature

Q35. Fine-tuning is more appropriate when the goal is primarily to:

A. Teach the model a desired behavior/style/task pattern
B. Perform real-time document retrieval
C. Replace a database
D. Search vectors

Section I — AI Agents

The JD specifically mentions **AI agents and agentic solutions using LangGraph, CrewAI and AutoGen**.

Q36. What is an AI agent?

A. A static database
B. A system that can use reasoning/planning and tools to accomplish a goal

C. A programming language
D. A vector embedding

Q37. What is the key difference between a basic LLM application and an agentic system?

A. An agent can decide actions/use tools as part of completing a goal
B. An LLM cannot generate text
C. An agent does not use models
D. There is no difference

Q38. What is tool calling?

A. Allowing an LLM/agent to invoke an external function or service
B. Calling a developer on the phone
C. Training a vector database
D. Creating embeddings manually

Q39. Which is a good example of an agent using tools?

A. LLM generates a poem
B. Agent searches a database, calls an API, performs calculations and produces a final answer
C. SQL database stores a row
D. User opens a browser

Q40. Why might an agent need memory/state?

A. To maintain relevant information across steps of a workflow
B. To increase monitor resolution
C. To replace the LLM
D. To eliminate APIs

Section J — Agentic Workflow

Q41. Consider:

User

↓

Agent

↓

Search Tool

↓

Database

↓

Agent

↓

Final Answer

What is happening?

A. An agent is using external tools as part of completing the task
B. A traditional SQL-only workflow
C. Model training
D. Static document retrieval only

Q42. When is a simple LLM call preferable to an AI agent?

A. When the task is straightforward and doesn't require planning or tool use
B. Always
C. Never
D. Only when databases are unavailable

This is an important conceptual point: **don't use agents just because they're fashionable.**

Section K — AI-Assisted Development

The JD specifically names **Claude Code, GitHub Copilot and Cursor** and expects AI-assisted development practices.

Q43. What is a major benefit of AI coding assistants?

A. They can accelerate coding, testing, debugging and documentation
B. They eliminate the need for developers
C. They guarantee bug-free code
D. They eliminate code review

Q44. An AI coding assistant generates code that passes one sample test but fails hidden edge cases. What should the developer do?

A. Deploy immediately
B. Review the code, identify edge cases and add tests
C. Trust the AI
D. Delete the project

Q45. Which statement best represents responsible AI-assisted development?

A. AI-generated code should be reviewed, tested and validated by developers
B. AI-generated code is always correct
C. Testing is unnecessary
D. Code review becomes unnecessary

Q46. How can GenAI assist software testing?

A. Generate test cases
B. Identify potential edge cases
C. Generate unit-test code
D. All of the above

Section L — AI + Software Engineering

The JD explicitly says AI should be applied across the **Application Development Lifecycle (ADLC)** to improve productivity, quality, testing and operational efficiency.

Q47. At which stages can AI coding assistants be useful?

A. Requirements
B. Coding

C. Testing
D. Debugging
E. Documentation
F. All of the above

Q48. An AI assistant suggests an API implementation. What should the developer verify?

A. Correctness
B. Security
C. Performance
D. Maintainability
E. All of the above